

Yipeng (Perry) Zhu

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Technical Skills

- **Cloud Engineering:** AWS, Kubernetes, Terraform, Ansible, Docker, Grafana, OpenTelemetry, GitHub Actions
- **Software Engineering:** TypeScript, Go, Python, Java, PostgreSQL, MongoDB, Spring Boot, NestJS, React

Work Experience

The University of British Columbia, Department of Computer Science

Teaching Assistant – CPSC330 Applied Machine Learning

Sep 2026 – Present

Samsung R&D Canada

Cloud Operations & Database Engineer

Sep 2025 – Aug 2026

- Designed and built DB-UAM, a centralized portal for secure and auditable database-user lifecycle management across MongoDB, MySQL, PostgreSQL, and Amazon Redshift. Deployed it to AWS EKS through Argo CD using Helm charts and provisioned dedicated subnets, a Karpenter NodePool, cross-account networking, VPC peering, IAM policies, and security groups with Terraform, supporting 3,000+ accounts across 150+ databases in multiple VPCs, AWS accounts, and regions.
- Migrated database observability from Datadog to a self-hosted Grafana LGTM stack integrated with AlertNow and Slack, using Ansible to deploy and configure Alloy across 80+ MongoDB EC2 instances for standardized metrics and logs. Customized a fork of an open-source MongoDB Prometheus exporter and configured an RDS exporter to support internal monitoring requirements and additional functionality, built MongoDB and RDS dashboards and alerts, and developed custom scripts for MongoDB index-usage analytics.
- Deployed MongoDB Ops Manager to AWS EKS using Helm charts and Terraform; configured TLS-encrypted communication, backup and recovery, high availability, and SSO integration with Authentik for secure and resilient MongoDB administration.

UBC Pacific Laboratory for Artificial Intelligence (PLAI)

Software Engineer – Infrastructure & Custom Development

May 2025 – Aug 2025

- Modernized legacy Java Minecraft plugins and mods from Minecraft 1.19 to 1.21 and refactored connection and session-lifecycle logic, improving the reliability of behavioral-data collection.
- Architected an Agent Companion Framework with session queuing and a command interface for private agent-player interactions; added deterministic safety guardrails to prevent unsafe in-game actions.
- Implemented the new PLAITime backend with context-aware voice transcription using Whisper, real-time LLM validation through AWS Lambda and UDP sockets, and an incentive mechanism that adjusted playtime based on interaction quality.

The University of British Columbia, Department of Computer Science

Teaching Assistant – CPSC304 Introduction to Relational Databases

Jan 2025 – Apr 2025

- Led weekly labs for 50+ students on database design, ER modeling, relational algebra, and SQL; mentored five teams on full-stack database projects.

Research & Publications

ClawBench: Can AI Agents Complete Everyday Online Tasks?

Yuxuan Zhang, Yubo Wang, Yipeng Zhu, et al. EMNLP 2026 Findings · COLM 2026 Workshop on Agent Behavior

- Architected and implemented ClawBench's core containerized runtime for safe, reproducible browser-agent evaluation across 281 tasks on 163 live websites, with interchangeable browser runtimes and agent harnesses, final-request interception, and five-layer recording.
- Maintained the open-source project by establishing CI/CD and automated PyPI release pipelines, shipping new browser-runtime and harness integrations, reviewing community PRs, and resolving user-reported issues.

Education

University of British Columbia

Bachelor of Science, Combined Major in Computer Science and Statistics

Sep 2023 – Expected May 2027